CardioVision AI

Multi-Vessel Risk Assessment & Clinical Vector Evaluation

RESEARCH PROTOTYPE EVALUATION REPORT — NOT FOR CLINICAL DIAGNOSIS

Evaluation Date: Wed, Oct 14, 2026, 06:31 AM
Patient ID: CV-DEMO-002
Architecture: 57-Feature Four-Model Architecture
Active Mode: Standard Production Mode

PATIENT NAME	PATIENT ID	AGE / BIOLOGICAL SEX	HEIGHT / WEIGHT / BMI
Alex Morgan	CV-DEMO-002	58 yrs · Male	170 cm / 72 kg (24.9 kg/m²)

1. Cardiovascular Prediction Summary

Multi-vessel predictive classification derived from the 57-dimensional clinical feature vector.

OVERALL CAD	LAD ARTERY	LCX ARTERY	RCA ARTERY
91.7%	**79.0%**	**65.0%**	**51.7%**
CAD	Stenotic	Stenotic	Stenotic
cad_model.pkl	lad_model.pkl	lcx_model.pkl	rca_model.pkl
Threshold: 50.0%	Threshold: 50.0%	Threshold: 50.0%	Threshold: 50.0%
High Est. Prob (≥70%)	High Est. Prob (≥70%)	Moderate Est. Prob (50–69%)	Moderate Est. Prob (50–69%)

Model Estimation Notice: Values are model-estimated statistical classification probabilities, not clinically calibrated epidemiological risk scores or measured anatomical luminal stenosis narrowing percentages.

2. Clinical Input Measurements Summary

Patient clinical parameters from active evaluation session evaluated against informational reference benchmarks.

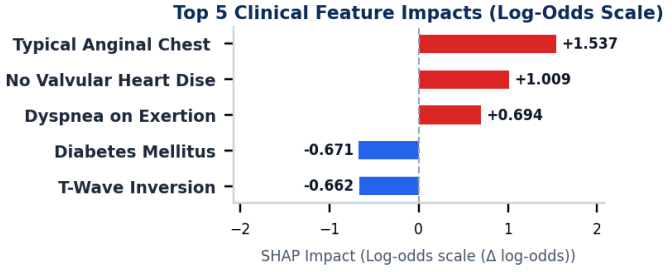
Clinical Parameter	Patient Value	General Reference Benchmark	Informational Evaluation Status
Patient Stature (Height)	**170 cm**	50 – 250 cm (Physiological reference)	Within adult stature reference
Body Mass Index (BMI)	**24.9 kg/m²**	18.5 – 24.9 kg/m² (WHO standard)	Normal weight reference (18.5 – 24.9 kg/m²)
Resting Systolic BP (BP)	**130 mmHg**	< 120 mmHg (resting systolic)	Above reference (≥ 130 mmHg, Informational)
Resting Pulse Rate (PR)	**70 bpm**	60 – 100 bpm	Within standard resting range
Left Ventricular EF (EF-TTE)	**50%**	≥ 50%	LVEF ≥ 50% (Standard systolic function)
Fasting Blood Sugar (FBS)	**98 mg/dL**	70 – 99 mg/dL	Within fasting range (70–99 mg/dL)
Serum Creatinine (CR)	**1.0 mg/dL**	0.7 – 1.3 mg/dL	Within displayed adult reference range (Informational)
LDL Cholesterol	**100 mg/dL**	< 100 mg/dL	Above standard reference (≥ 100 mg/dL, Informational)
HDL Cholesterol	**39 mg/dL**	> 40 mg/dL (M) / > 50 mg/dL (F)	Below reference (< 40 mg/dL, Informational)
Triglycerides (TG)	**122 mg/dL**	< 150 mg/dL	Within reference (< 150 mg/dL)
Evaluated Clinical Vector	**57 Model Inputs**	Standardized 7-domain vector	34 numeric + 23 binary clinical indicators synchronized

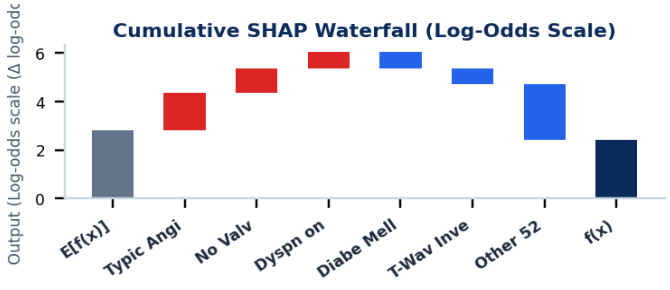
3. Patient-Specific SHAP Explainability

Feature contributions calculated via game-theoretic Shapley values. **Scale Note:** TreeExplainer outputs additive probability contributions (Δ probability). LinearExplainer outputs additive log-odds contributions (Δ log-odds). Scales are strictly separated.

CAD Model Explainability (cad_model.pkl) · Method: **LinearExplainer (Linear SHAP)** · Target: **CAD** · **[LOG-ODDS SCALE]**
Expected Baseline E[f(x)]: **2.810 log-odds** | Final Patient Output f(x): **2.407 log-odds** (σ(f(x)) = **91.7%** est. prob) | **✓ Additivity Verified: E[f(x)] + Σ ϕ_i = f(x)**

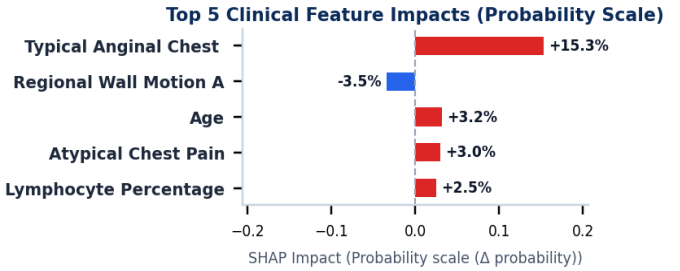
Rank	Clinical Feature	Category	Patient Value	SHAP (log-odds)	Impact Direction
#1	**Typical Anginal Chest Pain**	Symptoms	**Present (1)**	**+1.537 log-odds**	**Increases Risk**
#2	**No Valvular Heart Disease**	Valvular Disease	**Present (1)**	**+1.009 log-odds**	**Increases Risk**
#3	**Dyspnea on Exertion**	Symptoms	**Absent (0)**	**+0.694 log-odds**	**Increases Risk**
#4	**Diabetes Mellitus**	Medical History	**Absent (0)**	**-0.671 log-odds**	**Decreases Risk**
#5	**T-Wave Inversion**	ECG & Telemetry	**Absent (0)**	**-0.662 log-odds**	**Decreases Risk**

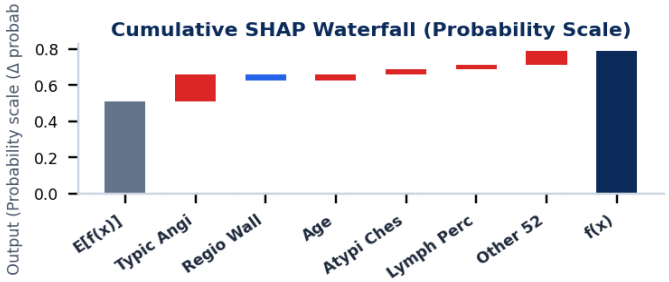




LAD Artery Model Explainability (lad_model.pkl) · Method: **TreeExplainer (Tree SHAP)** · Target: **Stenotic** · **[PROBABILITY SCALE]**
Expected Baseline E[f(x)]: **50.7%** | Final Patient Output f(x): **79.0%** | **✓ Additivity Verified: E[f(x)] + Σ ϕ_i = f(x)**

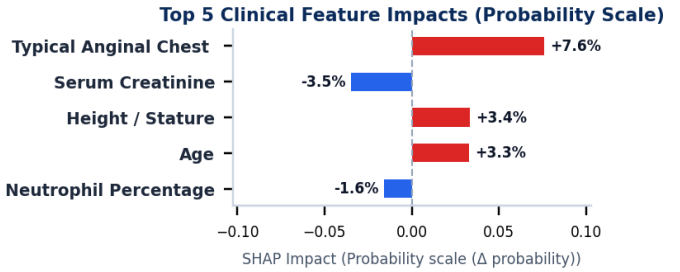
Rank	Clinical Feature	Category	Patient Value	SHAP (probability)	Impact Direction
#1	Typical Anginal Chest Pain	Symptoms	Present (1)	+15.3%	Increases Risk
#2	Regional Wall Motion Abnormality	Echocardiography	Absent (0)	-3.5%	Decreases Risk
#3	Age	Basic Demographics	58 years	+3.2%	Increases Risk
#4	Atypical Chest Pain	Symptoms	Absent (0)	+3.0%	Increases Risk
#5	Lymphocyte Percentage	Biomarkers	32 %	+2.5%	Increases Risk

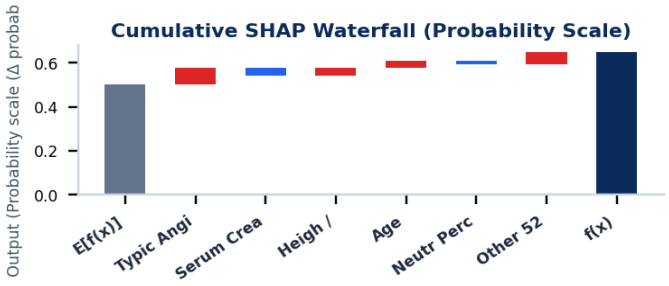




LCX Artery Model Explainability (lcx_model.pkl) · Method: **TreeExplainer (Tree SHAP)** · Target: **Stenotic** · **[PROBABILITY SCALE]**
Expected Baseline E[f(x)]: **50.2%** | Final Patient Output f(x): **65.0%** | **✓ Additivity Verified: E[f(x)] + Σ ϕ_i = f(x)**

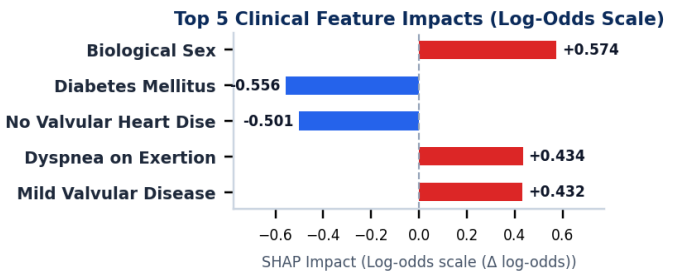
Rank	Clinical Feature	Category	Patient Value	SHAP (probability)	Impact Direction
#1	Typical Anginal Chest Pain	Symptoms	Present (1)	+7.6%	Increases Risk
#2	Serum Creatinine	Biomarkers	1 mg/dL	-3.5%	Decreases Risk
#3	Height / Stature	Basic Demographics	170 cm	+3.4%	Increases Risk
#4	Age	Basic Demographics	58 years	+3.3%	Increases Risk
#5	Neutrophil Percentage	Biomarkers	60 %	-1.6%	Decreases Risk

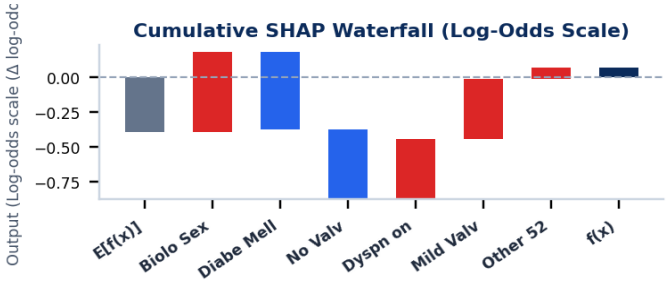




RCA Artery Model Explainability (rca_model.pkl) · Method: **LinearExplainer (Linear SHAP)** · Target: **Stenotic** · **[LOG-ODDS SCALE]**
Expected Baseline E[f(x)]: **-0.393 log-odds** | Final Patient Output f(x): **0.068 log-odds** (σ(f(x)) = **51.7%** est. prob) | **✓ Additivity Verified: E[f(x)] + Σ ϕ_i = f(x)**

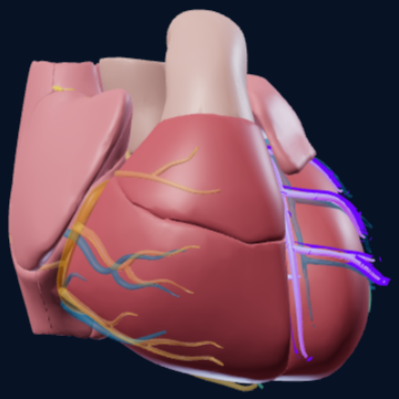
Rank	Clinical Feature	Category	Patient Value	SHAP (log-odds)	Impact Direction
#1	Biological Sex	Basic Demographics	Male	+0.574 log-odds	Increases Risk
#2	Diabetes Mellitus	Medical History	Absent (0)	-0.556 log-odds	Decreases Risk
#3	No Valvular Heart Disease	Valvular Disease	Present (1)	-0.501 log-odds	Decreases Risk
#4	Dyspnea on Exertion	Symptoms	Absent (0)	+0.434 log-odds	Increases Risk
#5	Mild Valvular Disease	Valvular Disease	Absent (0)	+0.432 log-odds	Increases Risk





4. 3D Coronary Heart Visualization

High-resolution anatomical perspective from the interactive Three.js 106-mesh BodyParts3D cardiac model.



Coronary Risk Telemetry

Evaluation Mode: **Standard Production Mode**

Spatial Anatomical Risk Overlay

CAD: 91.7% (CAD)
LAD: 79.0% (Stenotic)
LCX: 65.0% (Stenotic)
RCA: 51.7% (Stenotic)

Vascular Color Legend:

- LAD Arterial Trunk (#2a86ef)
- LCX Circumflex Trunk (#0fb77d)
- RCA Coronary Trunk (#ff9900)
- LM Coronary Stem (#e6d6c2)
- Ventricular Myocardium (#c94a53)

Anatomical Notice: 106-mesh BodyParts3D cardiac assembly; generic reference geometry.

5. Model Performance and Reliability

Standard Mode: Retrospective model evaluation metrics recorded during development on held-out test data.

Model	Active Model File	Target Vascular Outcome	Accuracy	F1 Score	ROC-AUC
CAD Model	cad_model.pkl	Overall Coronary Artery Disease	**85.2%**	89.7%	**0.917**
LAD Model	lad_model.pkl	Left Anterior Descending Stenosis	**65.6%**	67.7%	**0.756**
LCX Model	lcx_model.pkl	Left Circumflex Stenosis	**57.4%**	53.6%	0.584
RCA Model	rca_model.pkl	Right Coronary Artery Stenosis	**60.7%**	50.0%	0.652

Model Reliability Notice: Standard LCX and RCA baseline models exhibit limited discriminatory power in development cohort testing (ROC-AUC 0.584 and 0.652). Probabilities must be interpreted with extreme caution and never guide clinical interventions independently.

6. Research Limitations, Educational Guidance & Disclaimer

Scope of Research Evaluation & Clinical Considerations:

- Model estimates represent statistical pattern matching against retrospective cohorts, not definitive clinical diagnoses.
- Results must always be interpreted in full clinical context by a licensed medical practitioner.
- Machine learning probabilities cannot substitute for direct coronary angiography, CT angiography, or stress imaging.
- Diagnostic and management decisions depend on patient presentation, physical exam, and comprehensive clinical history.

RESEARCH PROTOTYPE — NOT A CERTIFIED MEDICAL DEVICE:

This software and report are intended strictly for educational, scientific research, and methodology evaluation purposes. They are not approved or cleared by any regulatory health agency (e.g., FDA, CE, PMDA) for clinical diagnostic or therapeutic decision-making. Predictions and SHAP explainability analyses do not substitute for formal cardiovascular imaging, laboratory testing, or physician consultation.